AP PRECALCULUS

Unit 1 • Lesson 1.11 • Guided notes

Topic 1.11 • Rational Functions and Concept of Limit

Guided notes • Fill in the blanks and work the examples in the boxes

Learning goals

- Use limit notation to describe the behavior of rational functions near holes, near vertical asymptotes, and at the ends.
- Rewrite rational expressions (factoring, polynomial division) to evaluate limits and find asymptotes.
- Use the remainder theorem and the binomial theorem to work with equivalent forms of polynomials.

Vocabulary. limit, one-sided limit, equivalent expression, polynomial division, remainder theorem, binomial theorem, Pascal's triangle.

1. Three kinds of limit statements

Key idea

Hole at $x = a$: $\lim_{x \to a} r(x) =$ _____ ($r(a)$ undefined)

Vertical asymptote $x = a$: $\lim_{x \to a^{\pm}} r(x) =$ _____

Horizontal asymptote $y = b$: $\lim_{x \to \pm\infty} r(x) =$ ____

Example 1 • Describe everything with limits

$r(x) = \dfrac{2x^2 + 2x - 4}{x^2 - 1}$.

2. Equivalent forms reveal limits

Key idea

Dividing $p(x)$ by $(x - a)$ gives $p(x) = (x - a)q(x) + R$, and the **remainder theorem** says $R =$ _____. So $(x - a)$ is a factor exactly when _____.

Writing $\dfrac{p(x)}{d(x)} = q(x) + \dfrac{R(x)}{d(x)}$ shows that the graph approaches $y = q(x)$ as $x \to \pm\infty$ when $\deg R < \deg d$.

Example 2 • Synthetic division

Divide $2x^3 - 3x^2 - 11x + 6$ by $x - 3$ and factor completely.

Example 3 • A limit and an asymptote from division

Let $r(x) = \dfrac{x^3 + 2x^2 - 5x + 1}{x - 1}$. Find $\lim_{x \to 1^+} r(x)$ and describe the end behavior.

3. The binomial theorem

The coefficients of $(a + b)^n$ are row n of _____: row 4 is _____.

Example 4

Expand $(x - 2)^4$ and $(2x + 1)^3$.

Common mistakes

- Writing $\lim_{x \to a} r(x) = \infty$ when the one-sided limits have opposite signs.
- Leaving out zero coefficients in synthetic division ($x^3 + 5$ is $1, 0, 0, 5$).

Check your understanding

1. The remainder when $x^4 - 3x + 1$ is divided by $x + 1$ is ____.
2. $\lim_{x \to 2} \dfrac{x^2 - 4}{x - 2} =$ ____. 3. The coefficient of x^2 in $(x + 3)^3$ is ____.